CS & IT

Discrete Mathematics

Set Theory and Algebra

DPP: 6

Q1 The set of all positive rational numbers forms an abelian group under the composition * defined by a * b = (ab)/2.
Which of the following is/are TRUE?
(A) The identity element is 2
(B) The inverse of a is 4/a.
(C) The inverse of 4 is 1
(D) The identity element is 1

Q2 Let r be the set of all real numbers and * is a binary operation defined by a * b = a + b + ab.
Which of the following is TRUE?
(A) Identity element is 0.
(B) The inverse of –1 is 1.
(C) The inverse of a is –a/(a + 1).
(D) R is not a group.

Q3 The set G = {0, 1, 2, 3, 4, 5} is a group with respect to addition modulo 6.
Which of the following is false?
(A) The inverse of 2 is 4
(B) The inverse of 3 is 3
(C) The inverse of 5 is 2
(D) The inverse of 1 is 5

Q4 G = {1, –1, i, –i}is a group w.r.t multiplication. The order –i is

Q5 If G is a group of order p, where p is a prime number. Then the number of sub groups of G is____.
(A) 1 (B) 2
(C) p –1 (D) p

Q6 Which of the following is True?
(A) The union of any two sub groups of a group G is also a sub group of G.
(B) The intersection of any two sub groups of a groups of a group G is also a sub group of G.
(C) Let G = {0, ± k, ± 2k, ±3k, ±4k, …..∞} where k is any fixed integer is a group w.r.t. addition.
(D) Every sub group an abelian group is also an abelian group.

Q7 Which of the following is True?
(A) Every cyclic group is an abelian group
(B) If 'a' is generator of a cyclic group then a^{-1} is also a generator of G.
(C) The order of a cyclic group is equal to the order of its generating element.
(D) A sub group of a cyclic group need not be cyclic.

Q8 How many generators are there for the cyclic group $(\{1, 2, 3, 4, 5, 6\}, \oplus_7)$

Q9 Which of the following is false?
(A) A cyclic group with only one generator can have at most 2 elements.
(B) The order of a cyclic group is equal to the order of its generator.
(C) The group $(\{1, 2, 3, 4\}, \otimes_5)$ is cyclic
(D) A group of order 4 is cyclic.

Q10 Let S = {0, 1, 2, 3, 4, 5, 6, 7} and * denotes multiplication modulo 8 that is x * y = (xy) mod 8.
Which of the following is a group w.r.t *?
(A) {0, 1} (B) {1, 4}
(C) {1, 3} (D) {1, 6}

Answer Key

Q1	(A, B, C)
Q2	(A, B, D)
Q3	(C)
Q4	4 to 4
Q5	(B)
Q6	(B, C, D)
Q7	(A, B, C)
Q8	2
Q9	(D)
Q10	(C)

Hints & Solutions

Q1 Text Solution:

Let e be the identity element.

$\therefore$ $a * e = a$

$\Rightarrow$ $(ae/2) = a$

$\Rightarrow$ $e = 2$

$\therefore$ Option (a) is true and option (d) is false.

Let a^{-1} = inverse of a

$a * a^{-1} = e$

$\Rightarrow \frac{a \times a^{-1}}{2} = 2$

$\Rightarrow a^{-1} = \frac{4}{a}$

Inverse of $4 = \frac{4}{4}$

$\therefore$ Option (b) and (c) are true.

Q2 Text Solution:

Let e be the identity element.

$\therefore$ $a * e = a$

$\Rightarrow$ $a + e + a.e = a$

$\Rightarrow$ $e = 0$

Let a^{-1} = inverse of a

$a * a^{-1} = e$

$a + a^{-1} + aa^{-1} = 0$ (\ 0 is identity element)

$\Rightarrow a^{-1} = \frac{-a}{a+1}$

$\therefore$ Inverse of –1 does not exist.

Hence, Option (b) is false.

Q3 Text Solution:

$5 \oplus_6 2 = 1$

$\Rightarrow$Inverse of 5 is not 2.

Q4 Text Solution:

Order of (–i) = 4, because the smallest integer n such that $(-i)^n = 1$ is n = 4

Q5 Text Solution:

Let (H, *) be a subgroup of order n, By Lagrange's theorem,

$\Rightarrow$ n is a divisor of p

$\Rightarrow$ n = 1 or n = p

$\Rightarrow$ H = {e} or H = G

$\therefore$ G has only 2 trivial subgroups

Q6 Text Solution:

G = {1, 3, 5, 7} is a group with respect to $\otimes_8$.

H_1 = {1, 3} and H_2 = {1, 5}

$H_1 \cup H_2$ = {1, 3, 5}

Here, H_1 and H_2 are subgroup of G,

But $H_1 \cup H_2$ is not a subgroup of G.

Q7 Text Solution:

Every sub group of a cycle group is cyclic (theorem)

Q8 Text Solution:

Number of generator of $G = \phi(6)$

= Number of positive integers which are less than 6 and relatively prime to 6.

= 2

Q9 Text Solution:

G = {1, 3, 5, 7} is a group with respect to $\otimes_8$.

G is not cyclic, because the generating element does not exist.

Q10 Text Solution:

(a) {0, 1} is not a sub group, because inverse of 0 does not exist.

(b) {1, 4} is not a sub group, because the set is not closed with respect to the given binary operation.

(c) {1, 3} is closed with respect to *.

$\therefore$ {1, 3} is a group.

(d) {1, 6} is not a sub group, because the set is not closed with respect to the given binary operation.